Digital Transformation
Understanding Business Goals, Risks, Processes, and Decisions

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Mathias Cöster, Mats Danielson, Love Ekenberg, Cecilia Gullberg, Gard Titlestad, Alf Westelius, and Gunnar Wettergren, *Digital Transformation: Understanding Business Goals, Risks, Processes, and Decisions*. Cambridge, UK: Open Book Publishers, 2023, https://doi.org/10.11647/OBP.0350

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ISBN Paperback: 978-1-80511-060-6
ISBN Hardback: 978-1-80511-061-3
ISBN Digital (PDF): 978-1-80511-062-0
ISBN Digital ebook (EPUB): 978-1-80511-063-7
ISBN XML: 978-1-80511-065-1
ISBN HTML: 978-1-80511-066-8
DOI: 10.11647/OBP.0350

Cover photo by Love Ekenberg
Cover design by Jeevanjot Kaur Nagpal

10. Globally Sustainable Digital Transformation

In this book, we have mostly covered digital transformation from a business perspective. But there is also a broader global sustainability perspective on transformation, as exemplified in this chapter by higher education. The Covid-19 pandemic accelerated the transformation and opened the world's eyes to understanding the possible impacts (both positive and not-so-positive) of transformation. For example, in the realm of higher education, one could compare the many universities in Sub-Saharan Africa that were forced to shut down overnight to the quick and successful shift to online teaching made by European universities.

In the UN policy brief "Leveraging Digital Technologies for Social Inclusion", a key message is: "Covid-19 is accelerating the pace of digital transformation. In so doing, it is opening the opportunities for advancing social progress and fostering social inclusion, while simultaneously exacerbating the risk of increased inequalities and exclusion of those who are not digitally connected."

Furthermore, the brief argues that the accelerated pace of digital transformation risks increasing the social exclusion of already vulnerable groups who are not digitally literate or connected.

UNESCO published a guide in 2021, "AI and Education: Guidance for Policy-makers". The authors state:

> However, while AI might have the potential to support the achievement of the Sustainable Development Goals (SDGs) of the United Nations, the rapid technological developments inevitably bring multiple risks and challenges, which have so far outpaced policy debates and regulatory frameworks. And, while the main worries might involve AI overpowering human agency, more imminent concerns involve AI's social and ethical

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> implications – such as the misuse of personal data and the possibility that AI might actually exacerbate rather than reduce existing inequalities.

AI is agnostic, in the sense that it can be implemented and used for both bad and good. AI is about technology, or a group of technologies that are normally implemented along with other technologies that increase their impact, for instance in learning analytics. AI and deep technologies can thus be seen as potential components of transformation, which ought to be carefully considered when preparing for change.

There is little consensus on how to define AI. UNESCO has used a pragmatic definition:

> AI might be defined as computer systems that have been designed to interact with the world through capabilities that we usually think of as human.

This definition implies that what we think of as AI will change over time. As our digital tools become more powerful and help us to solve more complex tasks, we get used to them, and start viewing them as normal automation rather than "intelligence". According to this perspective, AI is always at the frontline of automation; handling "intellectual" tasks that we previously had to rely on people to perform. This relativistic character of AI also tends to lead to blurred lines between AI and other technologies. For example, is learning analytics about AI, at least partly? The issue of intelligence also relates to philosophy. There is some dispute over how far the I in AI relates to the I in human intelligence, HI. Can AI be used without HI? The human decision component might be easy to underestimate in the context of AI. As with all other digitisation, this book argues that it would be unwise to believe that AI will provide value without mindful human consideration of when and how to apply it.

Observing transformations through AI in higher education and research, three serious gaps become visible:

- The first is the gap between expectations and reality. This is discussed in a 2022 EDUCAUSE report on AI. As is typical for IT-related products and concepts, AI might also eventually meet a winter of discontent when the hype surrounding it leads to disappointment and criticism in the face of little or no tangible benefits.

- The second gap is between developed countries and less developed countries, in particular in Sub-Saharan Africa. This is evidenced by the many shutdowns endured by many sectors in Sub-Saharan Africa during the Covid-19 pandemic, and the resulting setbacks they have caused. African countries are hardly visible in the statistics for research on AI (except for South Africa, which has 2000 AI publications). In the chart below, Sub-Saharan countries (black bubbles) show almost no research activities and very low GDP per capita. Figure 10.1 plots countries by the number of AI publications, GDP per capita, number of all scientific publications (bubble size), and region (colour).

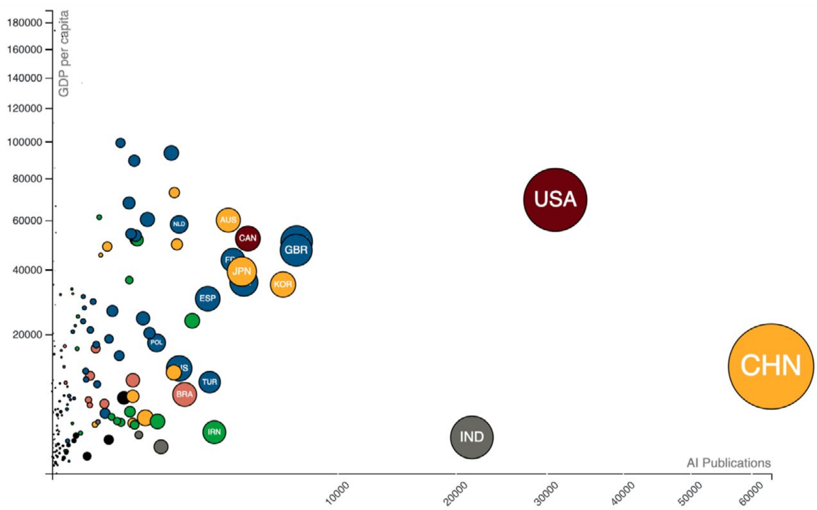

Figure 10.1. AI publications in Scopus vs GDP per capita by country and region.

- The third gap is between the business sector and higher education; the latter is lagging behind. This is detailed a bit more in the next section. If higher education, or certain parts thereof, is indeed lagging behind the business sector, it will fail to adequately prepare students for work life, and will increasingly find it difficult to interact with a more digitised business environment.

10.1. Digitisation in the Higher Education Sector

Higher education institutions and doctoral education can play a very important role in addressing the most burning issue of our time: sustainable development. Digitisation, if governed and implemented in an inclusive way, can play an important role in accelerating the positive impact by universities in meeting sustainable development goals. However, experience over the years shows that this is not a quick fix. It has taken time for universities to start to engage heavily with SDGs. The higher education sector has been slow to adapt to digitisation.

As mentioned in Chapter 1, for other parts of society, the Covid-19 pandemic accelerated the digital transformation of higher education, although most universities still have not started to prioritise digitisation projects. This means that whilst huge transformative processes have begun both for SDGs and digitisation, globally speaking we are still at the beginning. Research conducted by Boston Consulting Group shows that only about 30% of companies are successfully navigating digital transformation. There is no reason to believe that universities are doing any better.

For AI, the situation is in an even earlier phase. In 2022, EDUCAUSE found that very few higher-education institutions had implemented AI. Typically, only 1–4% were using AI for the twenty-four tasks examined, with three exceptions: proctoring (6%), plagiarism detection (20%) and chatbots/digital assistants (11%). For about twenty tasks, more than 40% of the institutions had no plan to use AI at all. However, quite a large share of planned projects is in the "tracking potential—planning/piloting" mode, which is typical (and perhaps sensible) for new areas of investment.

10.2. Need for a Higher Education Process Framework

Transformation in general is difficult, and digitisation is even more so. As a result, sound frameworks are needed to support higher education institutions and the sector in their digitisation processes.

In the study "Digital Transformation in Higher Education: A Framework for Maturity Assessment" (see Reading Tips), researchers explored and recommended a framework for maturity assessment, based on experiences from the United Arab Emirates, an advanced country driving digital transformation. The framework is based on the consulting company Deloitte's 2019 digital transformation assessment framework. This measures four processes: learning and teaching; enabling, e.g. library services; research and planning; and governance. Compared with the missions of the knowledge square (see Figure 10.3 below), one could ask if the innovations and services to society are underestimated in this model.

Researchers also mapped certain challenges for digitisation in higher education: (1) holistic vision, (2) staff competencies and IT skills, (3) data structure, data processing, and data reporting, (4) redundant systems, (5) third-party reporting systems, (6) manual entries, e.g. middle man, and (7) potential use by customers. This list shows the most important challenges (identified by 28-78% of the respondents). The EDUCAUSE special report on digital transformation in 2021 emphasises four barriers: (1) antiquated and siloed technology ecosystem, (2) lack of technology governance, (3) lack of necessary skills, and (4) change management difficulties.

These two findings from UAE and EDUCAUSE are similar but not the same and illustrate the necessity to carefully map maturity and challenges before starting or re-starting a digitisation process. A transformation must start with a thorough understanding of where the organisation is at, what the transformation is intended to achieve, and realistic planning for how to move towards the intended goals.

To successfully reap the benefits and manage risks and challenges, a framework for digitisation is helpful. This applies both when considering a certain university function that is being digitally transformed "end-to-end", for example online education, and when the university takes on a comprehensive transformation, considering all four of its missions: education, research, innovation, and services to the society.

A framework can be roughly divided into four phases (see Figure 10.2), however since digitisation is not a finite process, it is shown as a circle. Digitisation is not a destination; it is rather a permanent state of evolution.

Figure 10.2. Framework phases (inspired by Slidebooks consulting).

While many top-rated consulting companies offer frameworks for digitisation, including maturity assessments, few details are publicly available, probably because this is a revenue-generating part of the companies´ business. Little research exists on frameworks for higher education. In the article "Deep Dive into Digital Transformation in Higher Education Institutions", Mamdouh Alenezi discusses seven existing consulting companies' models for digital transformation in higher education institutions. In addition, it discusses the KPMG, Microsoft, and Google frameworks in more detail. To facilitate digital transformation in higher education, the paper suggests focusing on poor prioritisation, decentralised decision-making, internal resistance, digital literacy of the faculty, and a narrow view on return on investments. The author thus subscribes to a belief in centralised, forceful transformation efforts imposed on the institution, and the fact that the benefits of such transformation may be difficult to determine monetarily. This mindset is beneficial to consulting companies aiming to sell digital transformation projects to higher-education institutions.

A comprehensive digitisation process framework for higher education requires a holistic approach that takes all aspects of the university mission (here illustrated as education, research, innovation, and service to society, the so-called knowledge square) into consideration.

Extra challenges arise for institutions that completely or partly commit to open operations, here understood as open education resources (OER), open science and open innovation. "Open" implies a broad interaction with all members of society and might, but need not be, free of charge. It could also denote access that is restricted to a (large) collaborating network. For resource-poor countries, inclusion in "open" networks could be highly valuable, but it is increasingly likely that openness presupposes digital access. Some degree of digital transformation is thus a prerequisite for interacting with and benefiting from other institutions' open initiatives.

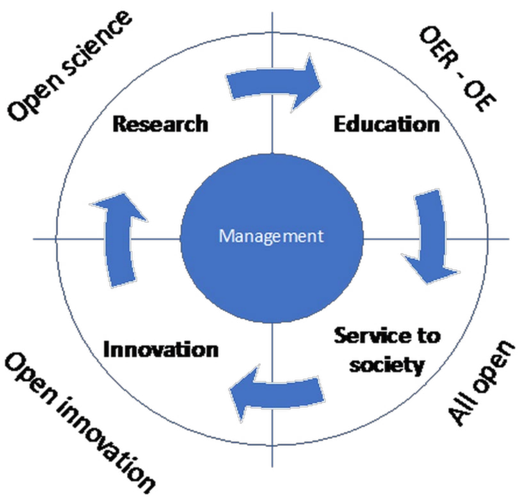

Figure 10.3. The knowledge square and open processes.

10.3. Aligning Higher Education Frameworks with SDGs

As discussed earlier in this chapter, digitisation efforts and policies have accelerated, and it is now one of the top twelve issues on the UN 2030 agenda. The importance of digitising higher education to meet the SDGs is widely acknowledged.

With the knowledge square in mind, universities have several tasks in relation to digitisation and SDGs. These include conducting research, building knowledge, and contributing to the global agenda for SDGs; educating students at all levels; empowering citizens to utilise digitisation and to engage with SDGs in their daily and professional lives; contributing to lifelong learning for all segments of society; and innovating on digitisation and how to better cope with the 2030 agenda.

The European Union gives high priority to digital transformation and sustainable development, both in its policies and programmes. The EU has launched the Digital Education Action Plan for 2021–2027, and the European Commission launched the Digital Decade Policy Programme in July 2022, stating in its press release:

> The Digital Decade policy programme is the way towards a more innovative, inclusive and sustainable future for Europe. Unlocking the potentials of the digital transformation, specifically by setting up and implementing multi-country projects, will pave the way for a competitive and sovereign Europe.

The Higher Education Sustainability Initiative (HESI), put both digitisation and the SDGs on their agenda at their 2022 Global Forum:

- Transformation of higher education post-Covid-19, including challenges and inequalities for those that lack capacities for rapid digital transformations and online learning.
- Integrating Sustainable Development Goals into higher education.

From a global perspective, digitisation requires missions to serve for the good, e.g., inclusion and sustainable development. Without such global missions, it is likely that individual higher education institutions will take a smaller, more inward-looking view of what their digitisation efforts should achieve. An important mission for higher education is expressed in SDG 4: to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The "for all" implies a need for global cooperation. The UN organised the Transforming Education Summit in September 2022. In its discussion paper on digital learning and transformation, three recommendations and three principles are suggested as a guide for digitisation. These recommendations and principles can be of use to all higher education institutions, whether they are embarking or re-embarking on digitisation processes.

	PRINCIPLES		
	PUT THE MOST MARGINALIZED AT THE CENTER	FREE, HIGH-QUALITY DIGITAL CONTENT	PEDAGOGICAL INNOVATION AND CHANGE
1. Ensure connectivity and digital learning opportunities for all	√	√	√
2. Build and maintain robust, free, public digital learning content and platforms	√	√	√
3. Focus on how technology can accelerate learning by enabling evidence-based instructional practice at scale	√	√	√

Figure 10.4. Recommendations and principles to guide digitisation in education.[1]

10.4. Chapter Summary

In this chapter, we have zoomed in, from a helicopter view on megatrend impact on the globe, identifying digitisation and AI as two of seven such trends – to some key issues for higher education institutions in meeting the challenges from an increasingly digitised world and at the same time observing the transformation of higher education to accelerate the impact of the SDGs while entering digital transformation. This is a tall order, but steps are increasingly being taken in this direction. The impact of digitisation has been accelerated by the Covid-19 pandemic, which also has made more visible three important gaps that need to be closed: between north and south, between expectations and reality and between the business sector and higher education.

Digitisation has in the last few years assumed a much more prominent place in work towards the sustainable development 2030 agenda. This is thanks to studies, influence and advocacy from many actors—and has not been without resistance. The Covid-19 pandemic was an eye-opener, which also accelerated policy surrounding digitisation. Now the leadership of global institutions such as the UN, UNESCO, and regional organisations such as the EU promote digitisation in all sectors through sector-specific and societal goals, including the SDGs.

1 Thematic Action Track 4 on 'Digital Learning and Transformation', Discussion paper (Final draft – 15 July 2022), https://transformingeducationsummit.sdg4education2030.org/AT4DiscussionPaper.

We have discussed how digitisation, AI, and deep technologies can be important for sustainability. And we have discussed how digitisation can be critical to the successful achievement of global higher education missions, and increase the impact of the SDGs. However, digital transformation is neither painless nor easy, and is not a destination; it is a permanent state of evolution. To better reap its benefits, meet its challenges and handle its risks, we have discussed the importance of using a framework for digitisation in higher education.

Several studies of such frameworks have been discussed, and we note that there is an obvious need for further research and development of holistic frameworks for higher education institutions to increase the success rates of digitisation processes.

We have also introduced some recommendations and principles for digitisation in education as a starting point for combining frameworks for the digitisation of higher education and sustainable development, and particularly SDG 4 for Education.

In Chapter 11, we will try to connect this chapter to earlier parts of the book, and will summarise its contents as a whole.

10.5. Reading Tips

Reports that provide a bird's eye view of what is happening globally, i.e., what trends to expect, the impact they might have, and where the "we" fits into this global picture, are always interesting. We recommend a report from Australia's National Science Agency:

- Naughtin C.; Hajkowicz S.; Schleiger E.; Bratanova A.; Cameron A.; Zamin T. and Dutta A. (2022). *Our Future World: Global Megatrends Impacting the Way We Live over Coming Decades.* Brisbane, Australia: CSIRO, https://www.csiro.au/en/research/technology-space/data/our-future-world.

The below report, with a forward-looking 2050 perspective, is helpful in shedding light on the (previous) lack of attention to digital transformation in the UN system and in work towards the SDGs.

- Nakicenovic, N. et al. (2019). *The Digital Revolution and Sustainable Development: Opportunities and Challenges*. Report prepared by The World in 2050 initiative. International

Institute for Applied Systems Analysis (IIASA), Laxenburg, Austria, https://doi.org/10.22022/TNT/05-2019.15913.

EDUCAUSE is an important open source of information, research and studies for those interested in higher education and technology, and particularly digital transformation. Here we recommend a few papers and publications that have been helpful in framing transformation and higher education.

- *EDUCAUSE Review: Special Report | Digital Transformation*, https://er.educause.edu/toc/educause-review-special-report-digital-transformation.
- *EDUCAUSE Review: Special Report Artificial Intelligence: Where Are We Now?* (2022), https://er.educause.edu/toc/educause-review-special-report-artificial-intelligence-where-are-we-now.
- McCormack, M. (2021, August 6). EDUCAUSE QuickPoll Results: Institutional Engagement in Digital Transformation. *EDUCAUSE Review*, https://er.educause.edu/articles/2021/8/educause-quickpoll-results-institutional-engagement-in-digital-transformation.

Harvard Business Review is also a good source for papers and articles on "practical digital transformation and AI", particularly for those with a business approach. Recommended reading includes:

- Davenport, T. H. and Redman, T. C. (2020, May 21). Digital Transformation Comes Down to Talent in 4 Key Areas. *Harvard Business Review*, https://hbr.org/2020/05/digital-transformation-comes-down-to-talent-in-4-key-areas.

Four academic papers are frequently used to focus on what global digital transformation is about, particularly in higher education. Gong and Ribiere clarify the definition of digital transformation.

- Gong, C. and Ribiere, V. (2021). Developing a unified definition of digital transformation. *Technovation* 102, 102217, https://doi.org/10.1016/j.technovation.2020.102217.

Alenezi, Holmström, and Marks et al. delve into frameworks, and their research can be helpful for those preparing to start (or restart) a transformation process.

- Alenezi, M. (2021). Deep Dive into Digital Transformation in Higher Education Institutions. *Education Sciences* 11 (12), p. 770, https://doi.org/10.3390/educsci11120770.
- Holmström, J. (2022). From AI to digital transformation: The AI readiness framework. *Business Horizons*, 65 (3), pp. 329–339, https://doi.org/10.1016/j.bushor.2021.03.006.
- Marks, A.; Al-Ali, M.; Atassi, R.; Abualkishik, A.Z. and Rezgu, Y. (2020). Digital Transformation in Higher Education: A Framework for Maturity Assessment. *International Journal of Advanced Computer Science and Applications* (*IJACSA*), 11 (12), https://doi.org/10.14569/IJACSA.2020.0111261.

Boston Consulting Group is one of the leading consulting companies offering valuable insights into digital transformation, AI, and deep technologies. Two of its reports on these topics are:

- Forth, P.; Reichert, T.; de Laubier, R. and Chakraborty, S. (2020). *Flipping the Odds of Digital Transformation Success*. Boston Consulting Group, BCG, https://www.bcg.com/en-nor/publications/2020/increasing-odds-of-success-in-digital-transformation.
- *What CEOs Need to Know About Deep Tech*. (2022, May 16). BCG Global, https://www.bcg.com/publications/2022/ceos-need-to-know-about-deep-technologies.

Studies of policy systems, policy development, and decisions are crucial for those wishing to understand what is happening and will probably happen in the world, and how they or their organisations can influence or take part in important development processes or fight against unwanted development. From a strategic perspective, it is always important to monitor and understand policy development relevant to one's organisation. Some see this as conducting intelligence to understand, interpret, and be prepared for what may come.

Here we suggest two reports that influence decision-makers. The Independent Group of Scientists report, issued every third or fourth year, is very important. We have also selected the DESA/UN policy brief as an example of precise policy work, which is read by politicians and civil servants.

- Independent Group of Scientists appointed by the Secretary-General. (2019). *Global Sustainable Development Report 2019 The Future is Now: Science for Achieving Sustainable Development*. United Nations, New York, https://sustainabledevelopment.un.org/gsdr2019.
- DISD/DPIDG. (2021). *Leveraging Digital Technologies for Social Inclusion* (Policy Brief UN/DESA Policy Brief #92:). United Nations Department of Economic and Social Affairs, https://www.un.org/development/desa/dpad/publication/un-desa-policy-brief-92-leveraging-digital-technologies-for-social-inclusion/.

The main messages from UN systems are normally issued by the Secretary General. Here, we have selected two documents relevant to digitisation and the SDGs. In addition, we also recommend the outcome document from the 2022 UN High-Level Political Forum on sustainable development.

- *Secretary-General's Report on "Our Common Agenda"* (2022), https://www.un.org/en/content/common-agenda-report/.
- *Secretary-General's Roadmap for Digital Cooperation* (2020), United Nations, https://www.un.org/en/content/digital-cooperation-roadmap/.
- *Outcome | High-Level Political Forum 2022* (2022), United Nations, https://hlpf.un.org/2022/outcome.

Some high-level forums can set goals for global cooperation, as in 2015, when certain heads of state and prime ministers first formulated the seventeen SDGs. More operational policies, such as the policy paper for the Transforming Education Summit Track 4 in September 2022, are developed at a lower level. It is good to understand how international cooperation paves the way for setting goals for digital transformation in education. UNESCO is the key UN agency for education (and science), and it decided to promote AI (both its opportunities and risks) and to offer policy advice early on. As a third reading suggestion, we have selected the UNESCO Institute for IT in Education. The IITE is a small but important operational entity under the UNESCO umbrella. Inclusive transformation for sustainable development is at the core of its strategy.

- *Thematic Action Track 4 on Digital Learning and Transformation Discussion Paper*, https://transformingeducationsummit.sdg4education2030.org/AT4DiscussionPaper.
- Miao, F.; Holmes, W.; Huang, R.; Zhang, H., and UNESCO. (2021). *AI and Education: Guidance for Policymakers*, https://unesdoc.unesco.org/ark:/48223/pf0000376709.
- UNESCO IITE. (2022). *UNESCO IITE Medium-Term Strategy for 2022-2025*. UNESCO IITE, https://iite.unesco.org/about-unesco-iite/.

In Europe, the European Union is setting the tone for policy development. If you are to operate in an EU environment, you definitely need to be aware of the relevant policies and developments in your area. For our purposes here, digital transformation, sustainable development, and higher education, two policy documents are selected for further reading, one providing a close perspective on education, and the other a broader EU scope.

- *Digital Education Action Plan (2021-2027) | European Education Area*, https://education.ec.europa.eu/node/1518.
- *The Digital Europe Programme | Shaping Europe's digital future*, https://digital-strategy.ec.europa.eu/en/activities/digital-programme.